M4-G RGBD Camera

MRDVS



The MRDVS M4-G is an iToF RGBD camera with a GMSL2 interface, designed for accurate, close-range perception in humanoid robots. It provides real-time depth sensing from 0.05 to 1 m, with accuracy as fine as 4 mm. Synchronized depth data and high-quality RGB images at 30 fps support close-range perception, tabletop grasping, and dexterous manipulation.

Key Features

1 Precise Depth Sensing

iToF provides higher accuracy and finer detail than stereo matching or speckle-based depth sensing.

2 Low-Light Performance

Active illumination delivers high-quality data in low light or at night.

3 Fewer Gaps in Depth Data

Active illumination reduces missing depth data for more complete depth maps.

4 Synchronized Data Output

Outputs depth and RGB images simultaneously, with RGB alignment support.

5 Reliable Long-Distance Transmission

GMSL2 supports longer cable runs and greater interference immunity than USB for stable data transmission.

Applications: Humanoid Robots

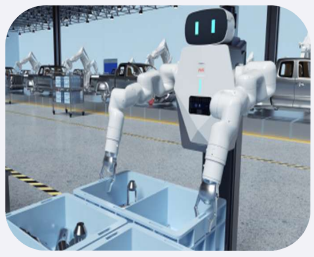
Tabletop Grasping

Wrist-mounted vision for VLA-based humanoid robots supports autonomous object recognition and grasping in home and office environments.

Dexterous Manipulation

Precise visual guidance supports industrial part recognition, grasping, and assembly, including bin picking, precision assembly, and machine tending.





400-025-6680 No. 7-902, China Artificial Intelligence Town, Hangzhou City Website: https://mrdvs.com/ More information: https://hub.mrdvs.cn/

M4-G RGBD Camera

Product Specifications

Model	M4-G		
Depth Technology	iToF	Power Consumption	3 W
Output Data	Depth / RGB / Point cloud	Laser Wavelength	940 nm
Depth Resolution / Frame Rate	640 × 480 (VGA) 30 fps (typical)	Dimensions	60 × 30 × 29 mm
Depth FOV (H × V)	90° × 70° (±3°)	Weight	< 200 g
RGB Resolution / Frame Rate	1280 × 1056 30 fps (typical)	Power Supply	12 V DC / 1 A
RGB FOV (H × V)	115° × 100° (±3°)	Data Interface	SerDes
RGB Pixel Size	2.5 µm	IP Rating	IP54
RGB ISP	Built-in	Operating Temperature	−20 °C to 60 °C
RGB HDR	≥ 100 dB	Storage Temperature	−40 °C to 85 °C
RGB Lens Aperture (F#)	2.05	Software Support	C / C++ SDK
RGB Focal Length (EFL)	1.28 mm	Supported Systems	Linux, Arm Linux
Measurement Range	0.05 to 1 m	Cooling	Passive cooling
Measurement Accuracy	1%	Eye Safety	Class 1

V2.1 2026-09-17

